

Hand-in assignment 1 solutions

Please note that these solutions are only meant to serve as a guide, the questions may have a number of alternative acceptable solutions. The advise is: **If you do not get full mark for a question, try to do it again independently after going through this document.**

Solutions

1. (a) True. Let (X, d) be both a quasi-metric space and pseudo-metric space, we immediately get (M2) and (M3). We also have $x = y \Rightarrow d(x, y) = 0$, now suppose $d(x, y) = 0$, by (PM2) we get $d(y, x) = 0$ and from (QM1) we can conclude that $x = y$.
- (b) False. Let X be a non-empty set and consider quasi-metric space $(P(X), d)$ with

$$d(A, B) = \begin{cases} 0 & \text{if } A \subseteq B; \\ 1 & \text{otherwise.} \end{cases}$$
 The sequence (A_n) given by $A_n = \emptyset$ for each $n \in \mathbb{N}$ right converges to both $\emptyset$ and X .
- (c) False. Consider metric space $(\mathbb{R}, d_E)$. For each $n \in \mathbb{N}$ the set $[\frac{1}{n}, 4]$ is closed, however union $\bigcup_{n \in \mathbb{N}} [\frac{1}{n}, 4] = (0, 4]$ is not closed.
- (d) True. Let (X, d) be a finite metric space, we then have that the set $\{d(x, y) \mid x \neq y\}$ is finite and so has a minimum. Let $m = \min\{d(x, y) \mid x \neq y\} > 0$. Now, suppose (x_n) is a convergent sequence (with limit $x \in X$). Let $\varepsilon = m$, then there exists N_ε such that if $n \geq N_\varepsilon$ we have $d(x_n, x) < \varepsilon = m$ and so $d(x_n, x) = 0$, that is $x_n = x$ for all $n \geq N_\varepsilon$, therefore our sequence (x_n) is eventually constant.
- (e) False. Consider metric space $(\mathbb{R}, d_E)$. Take $A = (0, 1)$ and $B = (1, 4)$, we have $A \cap B = \emptyset$ and so $\overline{A \cap B} = \emptyset$, on the other hand we have $\overline{A} \cap \overline{B} = [0, 1] \cap [1, 4] = \{1\}$.
2. (a) For each $x, y, z \in X$ we have: (Note that (M1) and (M2) follow immediately from the fact that d is a metric).
- (M1) If $x = y$ then $d^*(x, y) = \frac{d(x, y)}{1 + d(x, y)} = 0$ since $d(x, y) = 0$, conversely if $d^*(x, y) = 0$, we have $\frac{d(x, y)}{1 + d(x, y)} = 0$, and so $d(x, y) = 0$, thus $x = y$.
- (M2) $d^*(x, y) = \frac{d(x, y)}{1 + d(x, y)} = \frac{d(y, x)}{1 + d(y, x)} = d^*(y, x)$.
- (M3) First note that the function $f : [0, \infty) \rightarrow [0, \infty)$ defined by $f(t) = \frac{t}{1 + t}$ is differentiable on $(0, \infty)$ and $f'(t) = \frac{1}{(1 + t)^2} > 0$, moreover $f(0) = 0$ and if $t > 0$ then $f(t) > 0$. We have that this function is increasing.

Now we have:

$$\begin{aligned}d^*(x, z) &= \frac{d(x, z)}{1 + d(x, z)} \\&\leq \frac{d(x, y) + d(y, z)}{1 + d(x, y) + d(y, z)} \\&= \frac{d(x, y)}{1 + d(x, y) + d(y, z)} + \frac{d(y, z)}{1 + d(x, y) + d(y, z)} \\&\leq \frac{d(x, y)}{1 + d(x, y)} + \frac{d(y, z)}{1 + d(y, z)} \\&= d^*(x, y) + d^*(y, z)\end{aligned}$$

The inequality in the second line follows using triangular inequality for d and that f given above is increasing. The inequality in the second last line follows from the fact that $d(u, v)$ is non-negative for each $u, v \in X$.